University of St.Gallen

Population Aging and Bank Deposit Funding: The Role of Financial System Structure

St.Gallen, 18th March 2026

From insight to impact.

Executive Summary – Paper outline

1 Motivation & research gap

- Life-cycle theory predicts that as households age, borrowing declines while demand for safe retirement assets increases, leading to higher deposit-type savings and lower loan demand (Modigliani & Brumberg, 1954)
- Recent empirical studies find that a higher share of elderly population is associated with greater retail bank deposits (Doerr et al., 2022; Doerr & Kabas, 2019; Avramidis & Pennacchi, 2025)
- However, despite extensive research on financial system structures (e.g., Beck et al., 2000) and household portfolio choices, existing studies do not examine whether this relationship varies across countries with different financial systems

2 Hypotheses

- **H1:** The positive effect of population aging on retail deposit funding is stronger in bank-based financial systems than in market-based financial systems
- **H2:** A higher share of the very old population (85+) is associated with higher retail deposit funding, regardless of the financial system structure

3 Main regression model

$$DepositRatio_{i,c,t} = \beta_1 Age65 + \beta_2 FinStructure + \beta_3(Age65 \times FinStructure) + \beta_4 ReplacementRatio + \beta_5 Age85 + \gamma X + \theta Z + \alpha_i + \delta_t + \varepsilon$$

4 Preliminary findings

- The Age65 × FinStructure interaction acquires the expected negative sign in the FE model (6), providing **directional support for H1** – suggesting the aging effect on deposits is stronger in bank-based systems
- However, the **coefficient is statistically insignificant** (bootstrap p=0.281), meaning **H1 cannot be formally confirmed**

5 Considerations

- Potential multicollinearity issues when including pension system controls indicated by research by Scharfstein (2018); however, if not controlling for pension systems, regression model will have omitted variable bias
- Further avoidance of OVB due to inclusion of both macro- and bank-level controls

Motivation, research gap & hypotheses

Illustration

Explanation

- Life-cycle theory as per Modigliani and Brumberg (1954) predicts that as households age, borrowing declines
- Overall wealth increases before retirement and gradually decreases thereafter (net wealth decrease)

- At same time, demand for safe retirement assets increases, leading to higher deposit-type savings and lower loan demand
- This effect leads to a changed allocation of investments and counteracts (or pot. offsets) the declining overall wealth as per life cycle theory

Research gap

- Recent empirical studies find that a **higher share of elderly population** is associated with **greater retail bank deposits** (Doerr et al., 2022; Doerr & Kabas, 2019; Avramidis & Pennacchi, 2025)
- As these papers focus on one specific country by comparing states, it remains to be analyzed whether this effect can be i) observed **across different countries**, and ii) its impact can be explained by the **financial system structure**

Hypotheses

- **H1:** The positive effect of population aging on retail deposit funding is stronger in bank-based financial systems than in market-based financial systems
- **H2:** A higher share of the very old population (85+) is associated with higher retail deposit funding, regardless of the financial system structure

Sources: Modigliani & Brumberg, 1954; Doerr et al., 2022; Doerr & Kabas, 2019; Avramidis & Pennacchi, 2025; Levine, 2002; Beck et al., 2000

Overview of the economic intuition

H1

Older households have **lower risk tolerance** due to shorter investment horizons

→ Higher demand for **safe assets** as result

→ Households **rebalance portfolios** toward low-risk assets

→ *Our key research contribution*

Bank-based: Safe assets = bank deposits

→ In bank-based systems: Deposits increase

Market-based: Safe assets = govt bonds, bond ETFS, …

In market-based systems: Deposits increase less or not at all

H2

Very old households face estate planning; want to simplify portfolios, ensure liquidity for healthcare costs, make inheritance easier

→ Complex assets (e.g., stocks, funds, private investments, …) are gradually liquidated

→ Wealth moves into the simplest liquid accounts (i.e., bank deposits), regardless of financial system type

Sources: Modigliani & Brumberg, 1954; Doerr et al., 2022; Doerr & Kabas, 2019; Avramidis & Pennacchi, 2025; Levine, 2002; Beck et al., 2000

Our proposed regression model

FOR DISCUSSION

Regression component	Explanation	Expected sign
$DepositRatio_{i,c,t} = \beta_1 Age65_{c,t}$	Baseline aging effect	+
$+ \beta_2 FinStructure_{c,t}$	Financial system structure effect	-
$+ \beta_3 (Age65_{c,t} \times FinStructure_{c,t})$	Aging effect depending on structure **(H1)**	-
$+ \beta_4 ReplacementRatio_{c,t}$	Controls for pension system	-
$+ \beta_5 Age85_{c,t}$	Very old population effect **(H2)**	+
$+ \gamma X_{i,t}$ *See annex*	Bank-level controls + macro controls	*Not relevant*
$+ \theta Z_{c,t}$	Additional pension structure controls	*Not relevant*
$+ \alpha_i + \delta_t + \varepsilon_{i,c,t}$	Bank & year fixed effects; error term	*Not relevant*

Overview of steps & methodology

		What?	Why?
1	Full dataset	• Bank dataset as provided; this is the starting point for analysis	• Ensure comprehensive coverage before restricting sample to relevant bank types
2	Data filtering	• Restrict to savings banks; filter deposit ratio; address missing data	• Ensures deposit-driven banks with local footprint remain in sample
3	Variance decomposition	• Decompose within vs. between variation for key regressors	• Validate fixed-effects identification and assess time variation in explanatory variables
4a	Descriptive statistics	• Summarize distribution, central tendencies, and ranges of key variables (i.e., outliers)	• Understand sample structure and verify economic plausibility of variables
4b	Split dataset by financial system	• Compare variables across bank-based and market-based financial systems	• Explore heterogeneity and motivate interaction effect in regression model
4c	Correlation matrix	• Analyze pairwise correlations among key regressors	• Detect multicollinearity risks and validate identification of interaction terms
4d	Descriptive plots	• Visualize distributions, trends, and relationships across variables and system types	• Identify patterns, outliers, and preliminary relationships supporting empirical strategy
5	Preliminary regression results	• Estimate panel regressions with fixed effects and interaction terms	• Allows for initial testing of hypotheses

Pre-regression testing

Step 1 - SNL Dataset: banks for EU27 + CH

i Full Dataset
All bank types · 28 countries

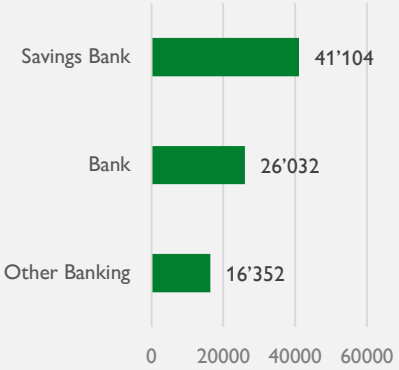
5,218
Banks

83,488
Observations

16 yrs
Time span

28
Countries

ii By Bank Type
3 categories breakdown

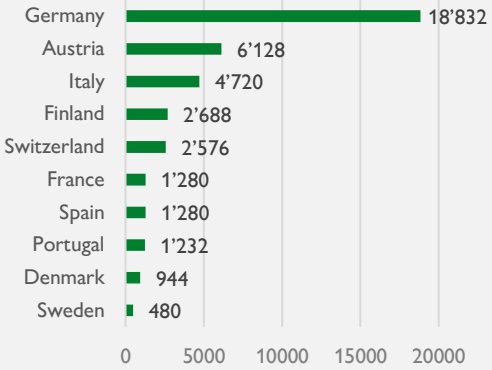


Share of total obs:
19.6% Other Banking
31.2% Bank
49.2% Savings Bank

iii Savings Banks
Deep-dive · 26 countries

2,569	41,104	26/28
Unique banks	Observations	Countries
Entity IDs	16.0 obs / bank	LT, RO absent

Observations by country (top 10 of 26)



⚠ Germany = 46% of obs · robustness check planned (excl. DE)

Step 2 - Dataset: data filtering & missingness

Sample Construction

Full dataset — 83,488
5,218 banks · 28 countries

▼

Savings banks only — 41,104
2,569 banks · 26 countries

▼

Deposit Ratio filtered — 31,439
Drops FY2025, and NAs in previous years
2,569 banks · 26 countries

Missingness Assessment

✓ Zero missingness in all core variables

Deposit Ratio · Total Deposits · Total Assets · Age65 · Pension dummies · ARR (4 missing, negligible)

FinStructure — only meaningful gaps

FinStructure	1,533 obs (4.9%)
Market cap / GDP	734 obs (2.3%)
Private credit / GDP	800 obs (2.5%)

Components are not independent — same country-year obs drive all three gaps.

Analytical Sample by Specification

Specification	Obs	Banks	Countries
Model 1 — no FinStructure	31,439	2,569	26
Models 2–5 — FinStructure	29,906	2,484	25
Δ attrition	*−1,533*	*−85*	*−1*

Step 3: Variance decomposition - within vs. between

Do key regressors move enough over time to survive fixed effects? All SDs computed on analytical sample (n = 31,439)

Variable	Overall SD	Between SD	Within SD	Within / Between
Deposit Ratio *Mostly cross-sectional — funding models stable YoY. Within SD still exploitable.*	0.136	0.127	0.047	0.371
Age65 (%) *Best result. Aging moves meaningfully within countries — FE identification validated.*	1.737	1.411	1.013	0.718
FinStructure *Weakest FE identification. Post-GFC convergence argument holds but expect wider SEs on β_3.*	0.175	0.200	0.072	0.359
Replacement Ratio (ARR) *Decent within-variation — pension reforms shifted ARR across 2010–2024.*	0.075	0.067	0.033	0.498

Takeaway No red flags — all variables have exploitable within-variation for FE estimation. *Age65 is best-identified (0.718). FinStructure is weakest (0.359) — flag this honestly and connect to the post-GFC convergence argument in your methodology section.*

Source: SNL Financial · World Bank WDI · Analytical sample: n = 31,439

Step 4a: Descriptive statistics

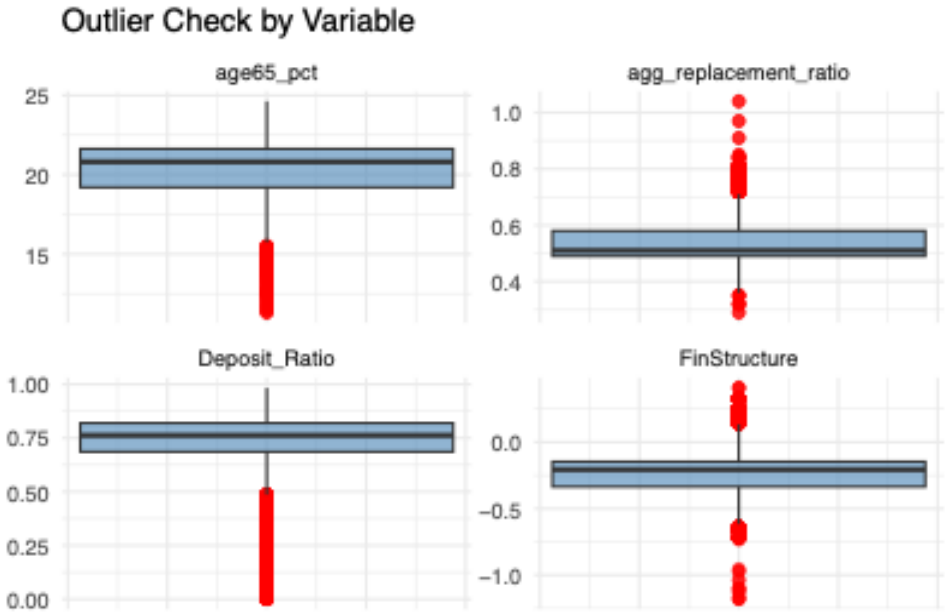
Analytical sample · Savings banks only · n = 31,439 obs · 2,569 banks · 26 countries

Variable	N	Mean	SD	Min	Median	Max	Note
Deposit Ratio	31,439	0.732	0.136	<0.01	0.762	0.983	*73% of assets funded via deposits — as expected for retail savings banks*
Age65 (%)	31,438	20.6	1.74	11.3	20.8	24.6	*Wide cross-country range. Within-variation (0.718) validates FE strategy ✓*
FinStructure	29,906	−0.211	0.175	−1.18	−0.207	+0.408	*Negative mean → sample is bank-based (Continental Europe). Weakest within-variation (0.359) — flag in methodology*
ARR	31,435	0.541	0.075	0.29	0.510	1.04	*54% avg. replacement rate. Post-GFC pension reforms drive within-variation (0.498)*
Bismarckian (dummy)	31,439	0.825	0.380	0	1	1	*82.5% of obs — reflects Germany's dominance (~49%). Structural, not sampling artifact*
Beveridgean (dummy)	31,439	0.105	0.307	0	0	1	*10.5% of obs. Denmark absent from Models 2–5 — flag as limitation*

Baseline category (both dummies = 0): Multipillar/Mixed — dominated by Switzerland (2,107 obs vs. Estonia 7, Latvia 4, Cyprus 22, Malta 49). *Consider recoding baseline to Beveridgean for easier economic interpretation — raise with professor.*

Step 4a: Descriptive statistics – Checking for outliers

Not really an issue; simply captures difference demographics across countries



Not really an issue; simply captures difference demographics across countries

! *Key question: Why are some deposit ratios so low? Potential for winsorizing (TBD)*

Strong methodology (Levine, 2002) as foundation

Step 4b: Descriptive statistics by financial system type

Motivation for H1 · Split at FinStructure = 0 · Analytical sample (excl. missing FinStructure)

Bank-based Market-based

Variable	Bank-based (n=26,540 · 25 countries)	Market-based (n=3,366 · 7 countries)	Comment
Deposit Ratio	0.730	0.738	**Δ +0.8 pp** *Market-based slightly higher — raw unconditional mean. H1 is about the aging interaction, not the level. Regression required.*
Age65 (%)	20.7%	19.6%	**Δ −1.1 pp** *Bank-based countries are older on avg. ⇒ aging & FinStructure correlated in cross-section. Watch for multicollinearity.*
FinStructure	−0.255	+0.136	**Split clean** *Binary classification at zero is meaningful — not driven by borderline cases.*
ARR	0.549	0.507	**Δ −4.2 pp** *Bank-based systems have more generous pensions (Bismarckian link). Watch ARR–FinStructure correlation in VIF diagnostics.*

⚠ *88% of obs are bank-based — market-based group (7 countries) is thin → wider SEs on market-based estimates.*

Takeaway No level difference in deposit ratios — this is exactly why the regression is necessary. H1 tests the aging–FinStructure **interaction**, which is invisible in unconditional means.

Step 4c: Correlation matrix, key regressors

Analytical sample · Pearson correlations · Pairwise complete observations

	Dep. Ratio	Age65	FinStr.	ARR	Bismarck.	Beveridg.
Dep. Ratio	—					
Age65	-0.060	—				
FinStr.	0.000	0.022	—			
ARR	-0.170	0.037	-0.421	—		
Bismarck.	-0.052	0.336	-0.680	0.247	—	
Beveridg.	-0.002	-0.104	0.362	-0.085	-0.745	—

Takeaway Age65×FinStructure ≈ 0 — interaction term well-identified. Flag ARR–FinStructure (−0.421) and Bismarckian–FinStructure (−0.680) collinearity. Report VIF after Model 3.

Key Findings

Bismarck. × FinStr. — 0.680
High collinearity — both capture the same Continental European institutional reality. Interpret in isolation with caution.

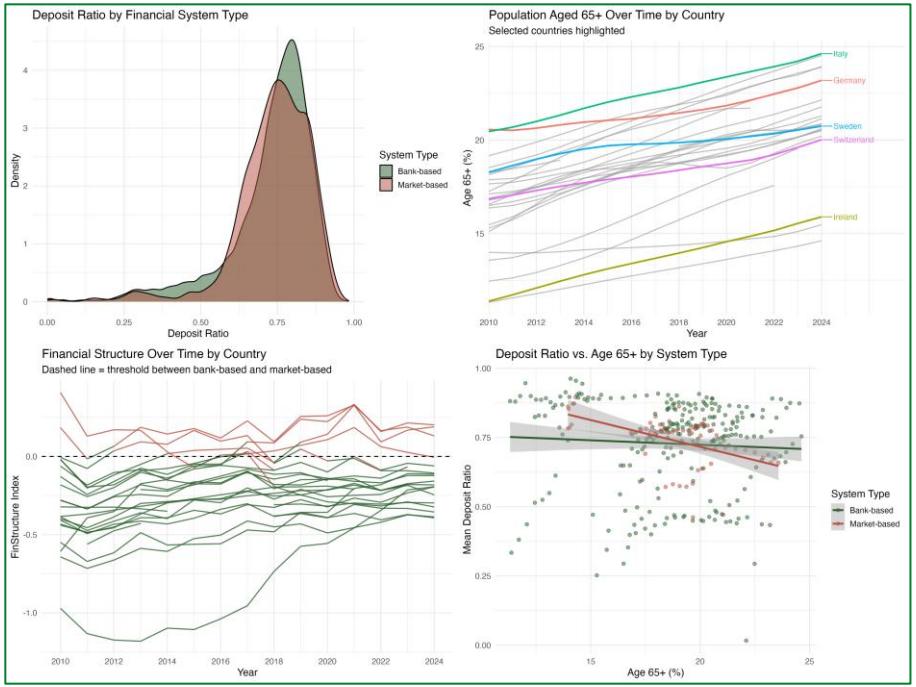
ARR × FinStr. — 0.421
Moderate collinearity — bank-based countries have more generous pensions (Scharfstein 2018). Run VIF after Model 3.

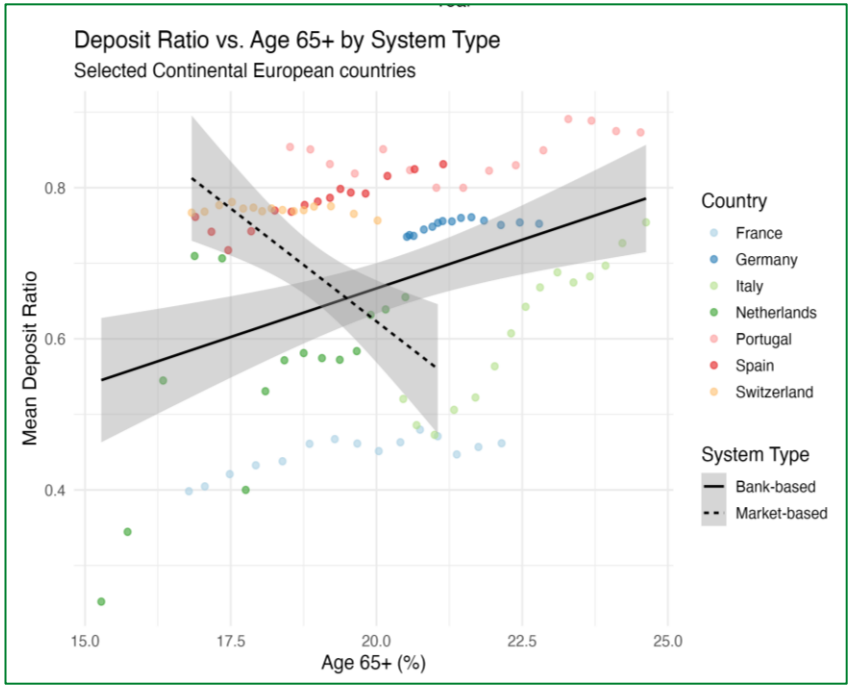
Beveridg. × Bismarck. — 0.745
Mechanical by construction (mutually exclusive dummies). Not a concern.

Age65 × FinStr. +0.022
Near zero — older countries are not systematically more bank-based. Interaction term β_3 is cleanly identified.

ARR × Age65 +0.037
Near zero — older countries do not have more generous pensions in this sample. Clean.

Step 4d: Descriptive plots





Regression output: preliminary findings (1/2)

Regression output: country-level clustering

Model:	Baseline (1)	+ FinStr (2)	+ Interact (3)	+ ARR (4)	+ Pension (5)	FE (6)
Variables						
Constant	0.829***	0.847***	0.779***	1.01***	0.913***	
	(0.163)	(0.189)	(0.209)	(0.199)	(0.179)	
Age 65+	-0.005	-0.006	-0.002	-0.004	0.003	-0.022
	(0.008)	(0.009)	(0.010)	(0.008)	(0.011)	(0.013)
FinStructure		0.001	-0.327	-0.112	-0.402	0.582*
		(0.114)	(0.869)	(0.782)	(0.778)	(0.289)
Age 65+ x FinStructure			0.017	0.002	0.013	-0.027
			(0.049)	(0.046)	(0.049)	(0.016)
ARR				-0.374	-0.373	0.343
				(0.289)	(0.292)	(0.217)
Bismarckian					-0.080	
					(0.106)	
Beveridgean					-0.049	
					(0.037)	
Fixed-effects						
Entity.ID						Yes
fiscal_year						Yes
Fit statistics						
Observations	31,438	29,906	29,906	29,903	29,903	29,896
R^2	0.00362	0.00474	0.00647	0.03948	0.04810	0.90185
Within R^2						0.09380
SE Type	Clustered	Clustered	Clustered	Clustered	Clustered	Clustered
Fixed Effects	No	No	No	No	No	Yes

Clustered (SP_COUNTRY_NAME) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*
Standard errors clustered at country level in parentheses throughout.

Interpretation

- **Age 65+**: negative sign unexpected given existing literature - implies ageing reduces rather than increases deposits. Statistically insignificant across all specifications (1-6)
- **FinStructure:** coefficient oscillates, and becomes positive in (6), and significant at 10% level. Not in line with expectations
- **Age 65+ * FinStructure**: interaction term only becomes negative in (6), providing directional support for H1. However, statistically insignificant – H1 cannot be confirmed
- **ARR:** turns positive in FE model (6), contrary to theoretical expectation of negative effect. Statistically insignificant (1-6)
- **R^2:** very low very low across pooled models (1-5). FE model (6) shows overall R^2 = 0.90, inflated by bank fixed effects. Within R^2 = 0.094 is the relevant metric - macro regressors explain approximately 9.4% of within-bank deposit variation

With **G=25** country clusters, asymptotic clustered standard errors are unreliable
→ **downward-biased SEs** and over-rejection of the null

Regression output: preliminary findings (2/2)

Regression output: wild-cluster bootstrap

Model:	Baseline (1)	+ FinStr (2)	+ Interact (3)	+ ARR (4)	+ Pension (5)	FE (6)
Variables						
Constant	0.829***	0.847***	0.779***	1.01***	0.913***	
	(0.163)	(0.189)	(0.209)	(0.199)	(0.179)	
Age 65+	-0.005	-0.006	-0.002	-0.004	0.003	-0.022
	(0.008)	(0.009)	(0.010)	(0.008)	(0.011)	(0.013)
FinStructure		0.001	-0.327	-0.112	-0.402	0.582*
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Age 65+ x FinStructure			0.017	0.002	0.013	-0.027
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ARR				-0.374	-0.373	0.343
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Bismarckian					-0.080	
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Beveridgean					-0.049	
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Fixed-effects						
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Fit statistics						
Observations	31,438	29,906	29,906	29,903	29,903	29,896
R^2	0.00362	0.00474	0.00647	0.03948	0.04810	0.90185
Within R^2						0.09380
SE Type	Clustered	Clustered	Clustered	Clustered	Clustered	Clustered
Fixed Effects	No	No	No	No	No	Yes
Boot p (Age65)						0.7685
Boot p (FinStructure)						0.2838
Boot p (Age65xFS)						0.2806
Boot p (ARR)						0.5096

Clustered (SP_COUNTRY_NAME) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*
Standard errors clustered at country level in parentheses. Wild cluster bootstrap p-values reported for Model 6 following Cameron, Gelbach & Miller (2008). Bootstrap uses Rademacher weights with 9,999 iterations.

Interpretation

- **Wild Bootstrapping:** wild cluster bootstrap fixes SE downward bias, by recalculating standard errors more conservatively
- **Outcome:** coefficients remain insignificant, confirming that the borderline significance of FinStructure in the asymptotic results was misleading, and that H1 is not confirmed

Expand sample → more countries increases G, directly improving bootstrap reliability and cluster-level identification

Exclude thinly covered countries → improves cluster balance and reduces noise from low-observation units

Revisit variable construction → slow-moving country-level regressors have insufficient within-variation; alternative measures with more time-series variation may improve identification

Overview of key challenges

FOR DISCUSSION

	1 Misspecification of the regression model	2 Small cluster problem	3 Low within-variance in key country-level regressors
Explanation	• Current FE setup mainly estimates how changes **within a country** over time relate to deposit ratios • But theory and your plot are much closer to how countries with different demographic structures and financial systems differ in deposit funding	• Inference is fragile because the number of country clusters is low (<50), making conventional clustered SEs unreliable • Asymptotic clustered standard errors are unreliable as they are downward-biased SEs, leading to an over-rejection of the null	• Low wit**hin-variance** in key country-level regressors (e.g., financial structure) weakens identification • Adding more controls can **absorb** the little variation that remains • This inflates SEs, worsens multicollinearity, and makes estimates unstable

Solution I

- Change the model specification (focus on between-country variation rather than within?)

Solution II

- Keep model, but expand the sample (increases number of clusters; more cross-sectional variation, more variation in financial structure)

Our key questions for discussion

Hypotheses & economic intuition:

- Is the economic intuition clear enough and sufficiently based on current state of research?
- Does the granularity of the research question and the hypotheses meet the expectations for the research seminar or do they need to be reformulated?

Regression model:

- There is a **low level of within-variance** for the explanatory variables. In an entity + time FE model, a lot of the between variance falls out. What is the right FE model specification?
- Due to **low number of country clusters**, there is insufficient within-variance, hence R raised a warning when including the macro controls. How should be best deal with this issue?
- How to deal with standard errors? We experience a small-cluster problem when simply clustering on a country level, and would hence propose to solve through **wild cluster bootstrapping as per Cameron, Gelbach, and Miller (2008)** – is that a valid approach?
- Does our proposed regression model consist of sensible controls to minimize OVB? Including the pension structure controls (e.g., Bismarckian) does not help, as in the FE it will fall out. Would it be better to include more countries to increase number of country clusters and variance?

Data:

- We decided to express „financial system structure“ through one continuous variable which we calculated based on established metrics from the literature. Is this argumentation for the metric to be used strong enough?
- Visual data plotting shows some outliers, especially for deposit ratio. Is winsorizing a valid approach here?

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Annex

From insight to impact.

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Literature review & papers

From insight to impact.

Academic papers supporting the theory (1/2)

	Household portfolios (Guiso et al.)	Life cycle model (Danziger et al.)	Household Finance (Campbell)	Financial systems
Topic	Theoretical & empirical analysis of household portfolio structure	Consumption Behavior of the Elderly in the Society	Theoretical & empirical analysis of household portfolio structure	Bank-based vs. market-based financial systems
Year	2002	1982	2006	2002
Journal	Book	Journal of Post Keynesian Economics	Journal of Finance	Journal of Financial Intermediation
Key findings	Older households systematically shift wealth toward safe assets	The elderly save more than the young people	Older households allocate a larger share of their portfolios to safe and liquid assets	No system is superior for economic growth
	Supports H1 on broad level	*Supports H1 specifically, indicates support for H2*	*Mainly supports H1*	*Helps to distinguish between systems (H1)*

Academic papers supporting the theory (2/2)

Strongly supporting & recent

	Population ageing & bank risk taking (Doerr et al.)	Banking on demography (Doerr & Kabas)	Demographics & structure of bank BS (Avramidis & Pennacchi)	Bank-based system growth (Beck et al.)
Topic	What are the impl. of an aging population for financial stability?	Impact of ageing population on banking sector	The Effects of an Aging Population on the Structure of Bank BS	Financial intermediary and private saving rates
Year	2022	2019	2025	2000
Journal	BIS Working Paper	N.a.	Journal of Money, Credit, Banking	Journal of Financial Economics
Key findings	Areas with more elderly people experience higher bank deposits	A 10% increase in the elderly population increases bank deposits by about 4.6%	Banks in older regions rely more on retail deposits and less on wholesale funding	Bank-based financial systems rely heavily on deposits as the main savings channel
	Strong support for H1	*Strong support for H1*	*Strong support for H1*	*Supports intuition behind H1 split by system*

Academic papers on pension systems & classification (1/2)

	Addio et al.	Scharfstein	Caloia et al.
Topic	Pension & household savings cross-country analysis	Pension Policy and the Financial System	Household savings & shocks to occupational pension funds
Year	2020	2018	2025
Journal	ECB Working Paper	Journal of Finance	Journal of Pension Economics & Finance
Key findings	Strong public pensions → lower need for private retirement savings	Countries with large private pension systems have larger capital mkts	When pension funds become weaker, HH increase private savings
	Supports to include pension system as control	*Makes strong argument to include pension system*	*Supports to include pension system as control*

Academic papers on pension systems & classification (2/2)

Duplicate

	Bottazzi, R., Jappelli, T., & Padula, M.	Alessie, R., Angelini, V., & van Santen, P.	Scharfstein, D. S.
Topic	Retirement expectations, pension reforms, and their impact on private wealth accumulation	Pension wealth and household savings in Europe: Evidence from SHARELIFE	Presidential Address: Pension policy and the financial system
Year	2006	2013	2018
Journal	Journal of Public Economics	European Economic Review	The Journal of Finance
Key findings	Pension generosity can substitute for private wealth especially among informed households	+1 EUR increase in pension reduces private wealth by -0.47 EUR, implying that pension substitutes savings	-) Higher pension replacement rates tend → more bank-based financial systems -) households hold more deposits as a share of wealth in more PAYGO-oriented system
	Supports using replacement rate as proxy for pension generosity	*Uses pension wealth as a measure which takes replacement rate into account*	*Direct link between financial systemstem & pension type*

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Reasoning for included (control-) variables

From insight to impact.

Financial system structure as the key explanatory mechanism

	Pension system structure	Financial system structure
Key question	• How much wealth needs to be accumulated for retirement? → *Affects and determines wealth accumulation*	• Once wealth is accumulated, where are these savings allocated? → *Determines if savings go to banks or markets*
Mechanism	• Determines retirement saving needs	• Determines savings intermediation channel
Link to bank deposits	Indirect	Direct
Limitations	• Does not determine asset allocation • Only indirectly affects bank deposit levels • Even if we know the levels of wealth accumulation depending on system, we still need to know where this wealth is allocated	• Different measures exist • Evolvement of systems over time?

FinStructure: Overview

$$\text{FinStructure} = \ln\left[(1 + \text{Market Cap} / \text{GDP}) / (1 + \text{Private Credit by Banks} / \text{GDP}) \right]$$

Interpretation

Positive value → Market-based financial system (market cap > private credit)
Negative value → Bank-based financial system (private credit > market cap)

	Countries	Valid Obs.	Mean	Std Dev	Min	Max
Base (WB only)	22	252	−0.26	0.24	−1.18	0.41
Patched	27	366	−0.22	0.24	−1.18	0.41

Source: World Bank WDI, ECB SDW, BIS Total Credit Statistics, Nasdaq OMX Stockholm | Formula: Levine (2002)

FinStructure: Comparison vs. Levine (2002)

- Strongly bank-based (< −0.2)
- Mildly bank-based (0 to −0.2)
- Market-based (> 0)
- No data available

Country	Base	Patched	Levine
Cyprus	-0.782	-0.782	-1.11
Portugal	-0.567	-0.511	-7.52
Estonia	N/A	-0.403	N/A
Greece	-0.419	-0.419	-1.02
Austria	-0.392	-0.392	-2.46
Italy	-0.431	-0.339	-1.45
Latvia	N/A	-0.336	N/A
Slovak Republic	-0.331	-0.331	N/A
Malta	-0.322	-0.322	N/A
Czechia	-0.304	-0.304	N/A
Slovenia	-0.283	-0.283	N/A
Spain	-0.280	-0.280	-1.29
Bulgaria	-0.263	-0.263	N/A
Lithuania	N/A	-0.253	N/A

Country	Base	Patched	Levine
Ireland	-0.209	-0.209	-0.03
Germany	-0.195	-0.195	-1.53
Hungary	-0.179	-0.179	N/A
Croatia	-0.152	-0.152	N/A
Romania	-0.150	-0.150	N/A
Poland	-0.131	-0.131	N/A
France	-0.097	-0.080	-1.42
Luxembourg	-0.041	-0.041	N/A
Finland	N/A	-0.008	-1.29
Belgium	0.053	0.013	-0.36
Netherlands	-0.115	0.036	-0.60
Sweden	N/A	0.056	-0.15
Switzerland	0.122	0.167	-0.71
Denmark	N/A	N/A	-0.62

Structure Size from Levine (2002); N/A = not in Levine's original sample | Sorted by patched mean FinStructure

Age65: Overview

Definition: Share of the total population aged 65 and above, expressed as a percentage of total population (World Bank, SP.POP.65UP.TO.ZS).

Rationale: *Following Doerr, Kabaš & Ongena (2024) and Avramidis & Pennacchi (2025), the 65+ threshold most robustly captures the relationship between demographic aging and bank retail deposit funding.*

28	420	28	18.71	19.01	11.32	24.62
Countries	Valid Obs.	NA's	Mean	Median	Min	Max

Coverage: All 28 countries (EU27 + CH) have exactly 15 valid observations (2010–2024). 2025 missing for all countries — not yet reported by World Bank.

Source: World Bank WDI (SP.POP.65UP.TO.ZS) | Period: 2010–2024 | EU-27 + Switzerland

Overview of proposed control variables

FOR DISCUSSION

	1 Pension system controls	2 Bank-level controls	3 Macro controls
Rationale	• Determine "how much" wealth needs to be accumulated	• Capture bank-factors impacting bank funding	• Capture macro-economic factors impacting bank funding structure
Controls	• **Replacement ratio:** Proxies how much households need to save privately for retirement • **Pension system type:** Classification of way a pension system is organized; captures structural organizational differences among systems	• **Bank size:** Large banks rely more on wholesale funding • **Loan ratio:** Banks with more loans need stable funding (usually comes from deposits) • **Equity ratio:** Higher equity = less risk; hence less stable deposit funding needed • **Profitability:** Profitable banks attract deposits due to stronger reputation	• **GDP growth rate:** During strong growth periods, households earn higher income and accumulate more savings • **GDP per capita:** Reflects wealth differences across countries • **Interest rate:** Higher interest rate on retail deposits is more attractive for depositors
	Pension-system determines directly how much savings are needed for retirement	*Bank-specific ratios/metrics impacting a banks funding mix*	*Macro-economic context in which the bank funding is embedded*

Controlling for pension systems: Pension replacement rate

Definition

The aggregate replacement ratio is gross median individual pension income of the population aged 65–74 relative to gross median individual earnings from work of the population aged 55–64, excluding other social benefits

Formula

$$\frac{\text{Median Gross Individual Pension (65–74)}}{\text{Median Gross Individual Earnings (50–59)}}$$

Datasets

- European Union: 2014-2025
  - https://ec.europa.eu/eurostat/databrowser/view/tespm100__custom_20473233/default/table
- OECD 2013
  - https://www.oecd.org/content/dam/oecd/en/publications/reports/2013/11/pensions-at-a-glance-2013_g1g3404e/pension_glance-2013-en.pdf
- OECD 2012
  - https://www.oecd.org/content/dam/oecd/en/publications/reports/2012/06/oecd-pensions-outlook-2012_g1g1aeb6/9789264169401-en.pdf
- OECD 2011
  - https://www.oecd.org/content/dam/oecd/en/publications/reports/2011/03/pensions-at-a-glance-2011_g1g12126/pension_glance-2011-en.pdf
- OECD 2009
  - https://www.oecd.org/content/dam/oecd/en/publications/reports/2009/09/pensions-at-a-glance-2009_g1ghab79/pension_glance-2009-en.pdf

Separating "where" and "how much"

	Pension system	
Key question	• How much wealth needs to be accumulated for retirement? → *Affects and determines wealth accumulation*	• How much wealth needs to be accumulated for retirement given public "generosity"? → *Affects and determines wealth accumulation*
Mechanism	• Determines retirement saving needs	• Determines retirement saving needs
Link to bank deposits	Indirect	Indirect

Measures

	Pension Assets / GDP	Replacement Ratio
Limitations	• Proxies the observable stock of funded retirement wealth in the economy • Reflects the outcome of asset allocation, not just retirement saving incentives • Multicollinearity threat (can move with broader financial development)	• Proxies how much households need to save privately for retirement. • Is closer to the underlying saving incentive • Helps separate how much households need to save from where those savings are held • Depending on focus might be harder to get (clean data for EU 2014-2025)

Aggregate Replacement Ratio: Overview

Definition: Gross median individual pension income of the population aged **65–74** relative to gross median individual earnings of the population aged **55–64**, excluding other social benefits.

ARR = *Median pension income (age 65–74) / Median earnings from work (age 55–64)*

Rationale

Scharfstein (2018) documents that higher replacement rates are associated with more bank-based systems and greater household deposit holdings. **Bottazzi, Jappelli & Padula (2006)** show pension generosity substitutes for private wealth accumulation, supporting its use as a proxy for pension system generosity.

28	433	15	0.54	0.53	0.29	1.04
Countries	Total Obs.	NA's	Mean	Median	Min	Max

Coverage: EU-27 + Switzerland | 2010–2025 | 16 years per country | Full series sourced from Eurostat (tespm100)

Pension System Classification: Overview

Bismarckian	Beveridgean	Multipillar / Mixed
Earnings-related, contribution-based. Benefits proportional to past earnings. Dominant in Continental & Southern Europe.	Flat-rate, universal basic pension. Tax-funded. Dominant in Nordic & Anglo-Saxon countries.	Combination of public mandatory + private funded pillars. Switzerland's 3-pillar model is the archetype.
AUT, BEL, BGR, HRV, CZE, FRA, DEU, GRC, HUN, ITA, LUX, POL, PRT, ROU, SVK, SVN, ESP	DNK, FIN, IRL, NLD, SWE	CYP, EST, LVA, LTU, MLT, CHE
Dummy = 1 | 17 countries	Dummy = 1 | 5 countries	Baseline (both dummies = 0) | 6 countries

Source: Ebbinghaus (2021); Werding (2003); OECD Pensions at a Glance (2024)

Step 2 - Dataset: key flags & commentary

Pattern 1 — Early-year gaps in smaller Eastern European markets

Czechia (FY2010–2017) · Slovakia (FY2015–2022) · Bulgaria (FY2012–2017) · Croatia (FY2010–2012)

Thin or undeveloped stock markets in the early post-GFC period — World Bank market cap data unavailable. Impact is minimal given small country sample sizes.

Pattern 2 — Structural gaps in larger markets

Denmark: all 15 years, 652 obs — drops entirely from Models 2–5. Beveridgean group shrinks 5 → 4 countries (loses DK, retains FI, SE, NL, IE) in all specifications including FinStructure.

Easy to find DE

Germany: FY2024 only (795 obs) — negligible given 15,292 total German observations. Caused by delayed World Bank data publication.

Attrition moving from Model 1 → Models 2–5 is entirely attributable to FinStructure gaps: Denmark (652), Germany FY2024 (795), minor Eastern European gaps. Non-random — will be noted explicitly in the data section.

⚑ Flag for professor discussion

Denmark's complete absence from FinStructure specifications should be raised explicitly. It is the only fully missing Beveridgean country, which could affect the precision of pension system interaction estimates.
Potential remedy: source Danish stock market capitalisation data from Nasdaq OMX Copenhagen or the Danish central bank — worth discussing before finalising the dataset.